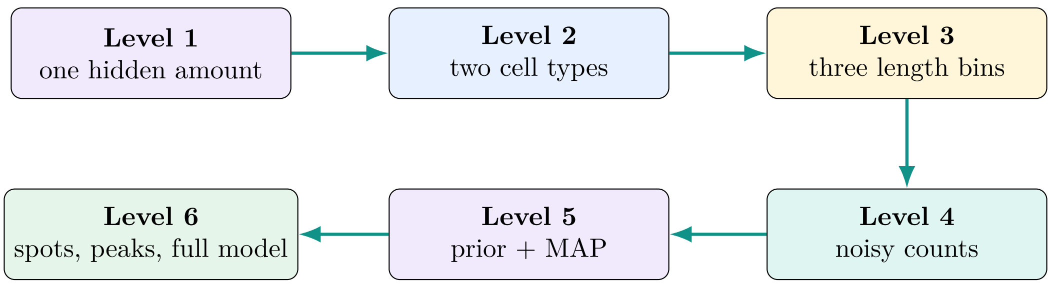
ShapeMix-ATAC, One Step at a Time

Start with one number; end with the full Bayesian model

A progressive tutorial with one running example and a new-symbol card for every new variable



Main idea

The tutorial keeps the same example all the way through: one spot contains a T-cell-like amount of 1.5 effective units and a B-cell-like amount of 0.5 units. Together they predict 24 cut sites, divided as 10 short-parent, 8 mono-parent, and 6 long-parent cut sites. We will not show the indexed version of the model until every part of this example makes sense.

Progressive rewrite, version 2.0

August 27, 2026

Contents

1 How to read this tutorial

The old way to present this model is to display the complete tensor equation and then define its symbols. This tutorial does the reverse:

1. explain a scientific object in ordinary language;
2. calculate it using the running numbers;
3. give it a short mathematical name; and
4. only later attach indices for many spots, peaks, bins, and cell types.

The colors show each quantity's job:

Observed measured data	**Fixed** reference input	**Unknown** inferred by MAP	**Calculated** from other values	**Output** reported result

Important distinction

The normative executable definition remains `docs/ShapeMix/model_specification.md`. This document explains that model; it does not replace the specification.

2 Before equations: what goes in and what comes out?

2.1 The output is a mixture, not a cell-type label

For each spatial spot, ShapeMix reports one row of estimated cell-type proportions. For example:

Spot	T cell	B cell	Monocyte
spot 1	0.65	0.25	0.10

The row sums to one. It means that normalizing the MAP abundance estimate gives approximately 65% T-cell-like, 25% B-cell-like, and 10% monocyte-like composition.

Important distinction

This row is not a confidence distribution. The current ShapeMix MVP reports one MAP point estimate. It does not report credible intervals, confidence intervals, or posterior samples.

2.2 The input counts cut sites

One deduplicated ATAC fragment has two Tn5 cut sites, one at each end. ShapeMix counts the cut sites that fall in selected peaks and labels each cut by the length of its parent fragment.

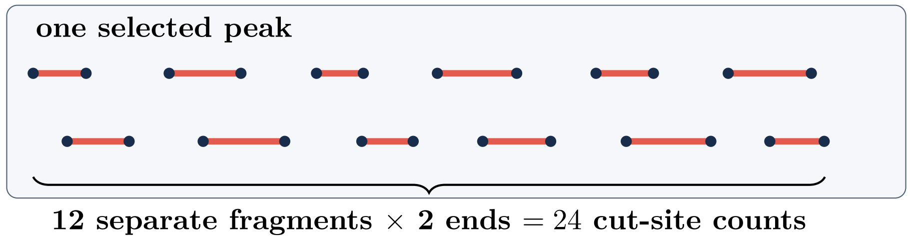


Figure 1: The fragment bars are separate observations. The two dark endpoints of each bar are the primary count units.

If both endpoints of all 12 fragments land in this peak, the peak total is 24. In real data a fragment can contribute zero, one, or two selected-peak cut sites, so bin counts need not be even.

Pause and check

Seven short, three mono-nucleosomal, and two long fragments have both ends in the same peak. How many cut-site counts are in the three bins?
Answer: 14 short, 6 mono, and 4 long; the total is 24.

3 Level 1: one hidden amount and one observed total

Imagine, temporarily, that only one cell type exists and we examine only one peak. A labeled reference cell typically contributes 12 cut sites to this peak. Our mixed spot contains 24 observed cut sites.

New symbol: A

Name: reference peak rate.
Job: a fixed input learned from labeled reference cells.
Units: expected cut sites per effective reference unit.
Running value: $A = 12$.

New symbol: z

Name: effective abundance.
Job: the positive hidden amount the model tries to infer.
Units: depth-scaled reference-cell-equivalent units, not literal nuclei.
Running value: try $z = 2$.

New symbol: μ, pronounced "mew"

Name: predicted mean count.
Job: the count predicted by a proposed hidden amount.
Units: expected cut sites.
Running value: $\mu = 2 \times 12 = 24$.

First write the relationship in words:

$$\text{predicted count} = \text{hidden amount} \times \text{known rate}.$$

Now insert the running numbers:

$$24 = 2 \times 12.$$

Only then write the compact equation:

$$\boxed{\mu = zA.}$$

New symbol: N

Name: observed peak total.
Job: the number actually measured in the spot and peak.
Units: integer cut sites.
Running value: $N = 24$.

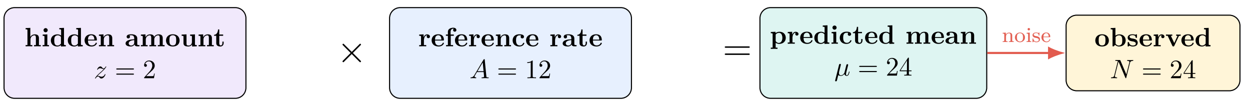


Figure 2: A proposed amount and a known rate produce a predicted mean; the experiment produces an observed count.

Pause and check

If $A = 5$ cut sites per unit and N is near 20, what effective abundance is a sensible first guess?
Answer: $z \approx 20/5 = 4$.

Symbol ledger after Level 1

Symbol	Meaning	Role
A	reference rate	fixed
z	effective abundance	unknown
μ	predicted mean count	calculated
N	observed total count	data

4 Level 2: two cell types, still one peak

Now allow T cells and B cells. We use the letters T and B as readable labels; the generic cell-type index will come much later.

$$A_T = 12, \qquad A_B = 12.$$

Try this hidden recipe:

$$z_T = 1.5, \qquad z_B = 0.5.$$

The predicted total becomes:

$$\boxed{\mu = z_T A_T + z_B A_B.}$$

With the running numbers:

$$\mu = 1.5(12) + 0.5(12) = 18 + 6 = 24.$$

Say the equation in words

Add the expected T-cell contribution and the expected B-cell contribution to get the expected peak total.

4.1 Why the total alone can be ambiguous

Because the two reference rates are equal, several recipes predict the same total:

z_T	z_B	predicted total μ
1.5	0.5	$1.5(12) + 0.5(12) = 24$
1.0	1.0	$1.0(12) + 1.0(12) = 24$
0.5	1.5	$0.5(12) + 1.5(12) = 24$

The peak total reveals that the combined effective amount is about two, but not which recipe produced it. This is the basic deconvolution problem.

New symbol: π_T and π_B

Name: normalized cell-type proportions.
Job: convert effective abundances into the reported mixture.
Range: between 0 and 1; together they sum to 1.

$$\pi_T = \frac{z_T}{z_T + z_B} = \frac{1.5}{2} = 0.75, \qquad \pi_B = \frac{0.5}{2} = 0.25.$$

Important distinction

$z_T = 1.5$ is not “one and a half nuclei.” It is a positive, depth-scaled reference-cell-equivalent amount. The normalized proportions π_T, π_B are the standardized biological output.

Pause and check

Why can $N = 24$ not distinguish the three recipes in the table?
Answer: $A_T = A_B$, so moving one unit from T to B leaves the predicted total unchanged.

5 Level 3: divide the same 24 cuts into length bins

The peak total hides how its cut sites were distributed by parent-fragment length. ShapeMix keeps three bins:

New symbol: L

Name: parent-fragment length.
Job: determines which length bin both endpoint cut sites inherit.
Units: base pairs (bp); later we calculate it as fragment end minus start.

Friendly name	Frozen length interval
short	$0 \le L < 100$ bp
mono (middle in figures)	$100 \le L < 250$ bp
long	$L \ge 250$ bp

5.1 First name the observed split

New symbol: $\boldsymbol{Y}$

Name: observed bin-count vector.
Job: records the short, mono, and long cut-site counts in that order.
Running value: $\boldsymbol{Y} = (10, 8, 6)$.

The bold type means "a list of numbers." Conservation gives:

$$N = 10 + 8 + 6 = 24.$$

Under the simplifying picture in which both ends of every fragment land in this peak, $(10, 8, 6)$ corresponds to 5 short, 4 mono, and 3 long fragments.

5.2 Then name the fixed reference shapes

New symbol: ω_T and ω_B

Name: conditional reference length fingerprints.
Job: for one cell type and one peak, give the expected fractions in the short, mono, and long bins.
Rule: each vector is fixed from training-reference cells and sums to 1.

Use:

$$\boldsymbol{\omega}_T = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right), \qquad \boldsymbol{\omega}_B = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right).$$

T-cell cuts are more often short-parent; B-cell cuts are more often long-parent.

5.3 Build the predicted bin counts

The T contribution is:

$$z_T A_T \boldsymbol{\omega}_T = 1.5(12)\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right) = (9, 6, 3).$$

The B contribution is:

$$z_B A_B \boldsymbol{\omega}_B = 0.5(12)\left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right) = (1, 2, 3).$$

New symbol: v

Name: predicted bin-count vector.
Job: add all cell-type contributions for each length bin.
Units: expected cut sites.

Running value: $\boldsymbol{v} = (10, 8, 6)$.

$$\boldsymbol{v} = (9, 6, 3) + (1, 2, 3) = (10, 8, 6).$$

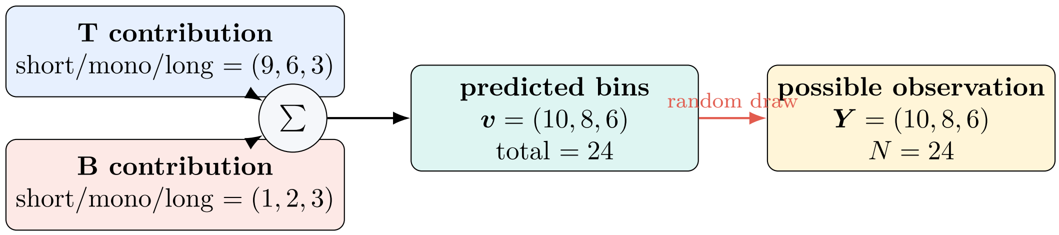


Figure 3: The predicted bin counts retain information that disappears when they are summed to 24.

New symbol: ρ, pronounced "row"

Name: predicted bin-probability vector.
Job: divide predicted bin counts by their total.
Rule: the entries are nonnegative and sum to 1.
Running value: $\boldsymbol{\rho} = (10, 8, 6)/24 = (5/12, 1/3, 1/4)$.

$$\boxed{\boldsymbol{\rho} = \frac{\boldsymbol{v}}{\mu}.}$$

5.4 The multinomial asks "how was the total split?"

After the total $N = 24$ is fixed, the conditional multinomial models how those same 24 cut sites are allocated:

$$\boxed{\boldsymbol{Y} \mid N \sim \text{Multinomial}(N, \boldsymbol{\rho}).}$$

The vertical bar means "given." The tilde means "is modeled as a random draw from." This is not 24 new observations: it is a second question about the same 24 cut sites.

Main idea

The total asks "how many?" The conditional shape term asks "how split?" When $A_T = A_B$, the total cannot distinguish T from B, but the short/mono/long split can.

Pause and check

What extra information does $\boldsymbol{Y} \mid N$ contain beyond N?
Answer: it records how the fixed total was divided among the three parent-length bins.

6 Level 4: observed counts are noisy

So far the observed values happened to equal the predictions. Real sequencing counts fluctuate. The model therefore distinguishes:

$$\underbrace{\mu}_{\text{predicted long-run center}} \quad \text{from} \quad \underbrace{N}_{\text{one observed total}}.$$

New symbol: e

Name: total effective abundance.
Job: add the effective amounts in the spot.
Running value: $e = z_T + z_B = 1.5 + 0.5 = 2$.

New symbol: ϕ_{ref}**, pronounced "fye ref"**

Name: reference inverse-dispersion.
Job: a fixed training-reference estimate controlling extra variability in peak totals.
Direction: larger inverse-dispersion means less extra variability.
Example value for this hand calculation: use $\phi_{\text{ref}} = 6$. This number is not hard-coded in ShapeMix.

Main idea

What "fixed" means here. ShapeMix first estimates ϕ_{ref} from the training-reference cells. Once that training step is complete, it holds the estimate fixed while inferring z for the mixed spots. A different dataset or training split can produce a different value. Version 1 freezes the *estimation procedure*, not the numerical value 6; the value 6 is used here only because it makes the arithmetic easy to follow.

Using the tutorial's example value, this spot's inverse-dispersion is

$$e\phi_{\text{ref}} = 2(6) = 12.$$

The peak total is modeled by a negative binomial:

$$\boxed{N \sim \text{NegBinomial}(\text{mean} = \mu,\ \text{inverse-dispersion} = e\phi_{\text{ref}})}\,.$$

Its variance is:

$$\text{Var}(N) = \mu + \frac{\mu^2}{e\phi_{\text{ref}}}.$$

With the running numbers:

$$\text{Var}(N) = 24 + \frac{24^2}{12} = 24 + 48 = 72.$$

Say the equation in words

Repeated versions of the same experiment would be centered around 24, but they would not all equal 24. The negative binomial allows more spread than a Poisson count with the same mean.

Important distinction

Some writing loosely calls the spot-level quantity $e\phi_{\text{ref}}$ "dispersion." In ShapeMix it behaves as an *inverse-dispersion*: increasing $e\phi_{\text{ref}}$ reduces the extra variance term $\mu^2/(e\phi_{\text{ref}})$.

Distribution	Question	ShapeMix role
Negative binomial	How many cut sites?	noisy peak total N
Conditional multinomial	How were they split?	bin vector $\boldsymbol{Y}$ given N
Gamma	Which positive amounts were plausible beforehand?	prior on z, introduced next

Pause and check

If $\mu = 24$, must N equal 24?
Answer: no. μ is the model's center; N is one noisy observation.

7 Level 5: prior, likelihood, posterior, and MAP

7.1 The fixed Gamma prior

Before examining the current spot, ShapeMix assigns the same positive prior to every cell type's effective abundance:

$$\boxed{z_T \sim \text{Gamma}(\text{shape} = 2, \text{rate} = 1), \qquad z_B \sim \text{Gamma}(\text{shape} = 2, \text{rate} = 1).}$$

This is the frozen version-1 prior. It has mean 2, variance 2, and mode 1.

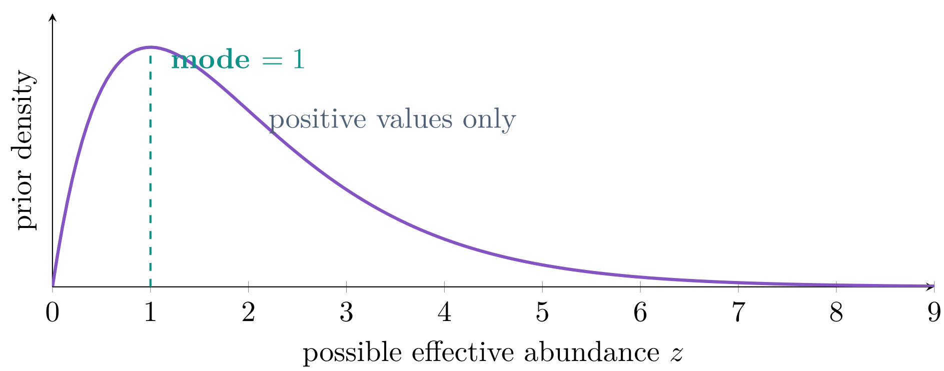


Figure 4: The Gamma$(2, 1)$ prior is a probability density over positive effective abundance.

Because T and B receive the same prior, no named type is favored. The prior does, however, prefer positive, moderate amounts over values extremely close to zero or extremely large values.

Important distinction

A prior is a modeling assumption, not a biological fact. Different plausible priors can yield different posteriors when the data are weak. The protocol freezes the prior before final evaluation and uses the same prior in both model arms; scientific sensitivity checks should disclose whether conclusions change under other predeclared priors.

7.2 How Bayes combines the pieces

Likelihood
How well does a recipe predict $N, \boldsymbol{Y}$?

Prior
Was that positive amount plausible beforehand?

$\propto$

Posterior
Updated support after seeing the data

In symbols:

$$\underbrace{p(z_T, z_B \mid N, \boldsymbol{Y})}_{\text{posterior}} \propto \underbrace{p(N, \boldsymbol{Y} \mid z_T, z_B)}_{\text{likelihood}} \underbrace{p(z_T)p(z_B)}_{\text{Gamma priors}}.$$

Here $p(\cdot)$ means a probability or probability density, the vertical bar means "given," and $\propto$ means "proportional to."

New symbol: $\widehat{\boldsymbol{z}}_{\mathrm{MAP}}$

Name: maximum-a-posteriori estimate.
Job: the single abundance vector at the highest point of the posterior.
Hat: a hat means "estimated."
MAP: maximum a posteriori, or "highest posterior."

$$\widehat{\boldsymbol{z}}_{\mathrm{MAP}} = \arg\max_{\boldsymbol{z}>0} \left[\log p(N, \boldsymbol{Y} \mid \boldsymbol{z}) + \log p(\boldsymbol{z})\right].$$

The software normalizes this best point to report:

$$\widehat{\pi}_T = \frac{\widehat{z}_T}{\widehat{z}_T + \widehat{z}_B}, \qquad \widehat{\pi}_B = \frac{\widehat{z}_B}{\widehat{z}_T + \widehat{z}_B}.$$

Important distinction

MAP reports the location of the posterior's highest point, not its width. Consequently, the MVP does not provide a confidence level or credible interval for each reported proportion. Optimization diagnostics are not uncertainty intervals.

Pause and check

Does a MAP estimate of 75% T mean "75% confidence that the spot is T"?
Answer: no. It is a point estimate of the spot's T-like mixture proportion.

8 Level 6: add peaks, spots, bins, and cell types

The small example used readable labels instead of general indices. We now add one index at a time because real data contain many peaks and spots.

8.1 First add multiple peaks

New symbol: p

Name: peak index.
Job: labels a selected genomic peak: $p = 1, 2, \ldots, P$.
Capital P**:** the number of selected peaks.

Adding p turns A_T into $A_{T,p}$ and μ into μ_p:

$$\mu_p = z_T A_{T,p} + z_B A_{B,p}.$$

Peak	$A_{T,p}$	$A_{B,p}$	Teaching purpose
1	12	12	total is ambiguous; shape can help
2	8	16	total itself helps distinguish types

For $z_T = 1.5, z_B = 0.5$:

$$\mu_1 = 24, \qquad \mu_2 = 1.5(8) + 0.5(16) = 20.$$

8.2 Then add multiple spots

New symbol: s

Name: spot index.
Job: labels a spatial spot: $s = 1, 2, \ldots, S$.
Capital S**:** the number of spots.

The mixture changes by spot, so z_T becomes $z_{s,T}$. Fixed reference fingerprints do not change by spot.

Spot	$z_{s,T}$	$z_{s,B}$	peak-1 predicted bins
spot 1	1.5	0.5	$(10, 8, 6)$
spot 2	0.5	1.5	$(6, 8, 10)$

Are the spot abundances estimated independently?

Yes—in ShapeMix version 1, each spot receives its own abundance vector and is scored using that spot's own observed counts. Spot 1's N, Y help estimate $(z_{1,T}, z_{1,B})$; spot 2's counts help estimate $(z_{2,T}, z_{2,B})$. There is no neighbor-to-neighbor abundance term.

Shared across all spots	Different for each spot
reference peak rates A	observed counts N, Y
reference length shapes ω	inferred abundances z
reference inverse-dispersion ϕ_{ref}	total effective abundance e
Gamma-prior parameters	predictions μ, ρ and output π
selected peaks and cell-type definitions	one separate reported mixture row

Important distinction

Software may process many spots together in one batch for speed. Batching is a computational detail; it does not make one spot's abundance affect another's. Spatial smoothing and hierarchical coupling are not part of version 1.

8.3 Finally replace names with general indices

New symbol: b and c

$b = 1, \ldots, B$ labels a parent-length bin; version 1 has $B = 3$.
$c = 1, \ldots, C$ labels a declared cell type; C is the number of types.
The earlier T and B labels are simply two possible values of c.

Index	Meaning
s	which spatial spot?
p	which selected genomic peak?
b	which parent-fragment length bin?
c	which reference cell type?

Examples:

- $Y_{3,12,2}$: observed cut-site count in spot 3, peak 12, bin 2.
- $A_{T,12}$: fixed T-reference total rate at peak 12.
- $\omega_{B,12,3}$: fixed B-reference long-bin fraction at peak 12.
- $z_{3,T}$: inferred T-like effective abundance in spot 3.

Important distinction

The italic subscript p labels a peak. The notation $p(\cdot)$ used in the Bayes section denotes a probability density. Context distinguishes them.

8.4 Array shapes before equations

Quantity	Dimensions	Role
z	$S \times C$	inferred spot-by-type amounts
A	$C \times P$	fixed type-by-peak rates
ω	$C \times P \times B$	fixed type-by-peak-by-bin fractions
N, μ	$S \times P$	observed and predicted peak totals
Y, v, ρ	$S \times P \times B$	observed counts, predicted counts, predicted fractions

9 The complete model, one line at a time

9.1 Line 1: total effective abundance

$$\boxed{e_s = \sum_c z_{s,c}.}$$

Say the equation in words

For one spot, add the effective amounts across all declared cell types.

9.2 Line 2: predicted count in one length bin

$$\boxed{v_{s,p,b} = \sum_c z_{s,c} A_{c,p} \omega_{c,p,b}.}$$

Say the equation in words

For one spot, peak, and length bin, calculate every cell type's contribution and add them.

$$\underbrace{z}_{\text{effective units}} \times \underbrace{A}_{\text{cuts/unit}} \times \underbrace{\omega}_{\text{fraction}} = \underbrace{v}_{\text{expected cuts}}.$$

9.3 Line 3: predicted total and bin fractions

$$\boxed{\mu_{s,p} = \sum_b v_{s,p,b}, \qquad \rho_{s,p,b} = \frac{v_{s,p,b}}{\mu_{s,p}}.}$$

Say the equation in words

Add the predicted bins to get the predicted peak total. Divide each predicted bin by that total to obtain fractions that sum to one.

9.4 Line 4: noisy observed peak total

New symbol: ϵ, pronounced "epsilon"

Name: numerical safety floor.
Job: a small fixed positive number that prevents invalid zero inverse-dispersion during computation. It is not inferred from the spot.

$$\boxed{N_{s,p} \sim \mathrm{NegBinomial}\big(\text{mean} = \mu_{s,p},\ \text{inverse-dispersion} = \max(e_s, \epsilon)\phi_{\mathrm{ref}}\big)\,.}$$

Say the equation in words

The observed peak total is a noisy count around the predicted mean. Larger spots receive the abundance-scaled inverse-dispersion required by the version-1 model.

9.5 Line 5: conditional length-bin split

$$\boxed{\boldsymbol{Y}_{s,p,\cdot} \mid N_{s,p} \sim \mathrm{Multinomial}\Big(N_{s,p}, \boldsymbol{\rho}_{s,p,\cdot}\Big)\,.}$$

The bold dot means "take the complete vector over all bins while holding s and p fixed."

Say the equation in words

Given the observed peak total, allocate those same cut sites among the three length bins according to the predicted fractions.

9.6 Line 6: prior and reported proportions

$$\boxed{z_{s,c} \sim \mathrm{Gamma}(2, 1), \qquad \pi_{s,c} = \frac{z_{s,c}}{e_s}.}$$

Say the equation in words

Give every cell type the same fixed positive prior, infer the MAP abundances, and normalize them so each spot's reported proportions sum to one.

9.7 Why the MAP calculation separates by spot

After all indices have been introduced, the MAP score can be written as a sum of one score for each spot:

$$\text{total MAP score} = \sum_s \text{spot score}_s,$$

$$\text{spot score}_s = \sum_p \big[\log p(N_{s,p} \mid z_s) + \log p(\boldsymbol{Y}_{s,p,\cdot} \mid N_{s,p}, z_s)\big] + \sum_c \log p(z_{s,c}).$$

The first sum measures how well z_s explains every peak in spot s; the last sum supplies that spot's Gamma-prior score. The fixed $A, \omega, \phi_{\mathrm{ref}}$ are shared by all spots and are left implicit in the probability notation.

No spot score contains another spot's abundance. Consequently, maximizing the total score is mathematically equivalent to maximizing each spot's score separately, conditional on the shared reference quantities. Each reported row is then normalized within its own spot:

$$\widehat{\pi}_{s,c} = \frac{\widehat{z}_{s,c}}{\sum_{c'} \widehat{z}_{s,c'}}.$$

Here c' is only a dummy summation label: the denominator adds the estimated abundances over every declared cell type in the same spot.

9.8 One generative diagram

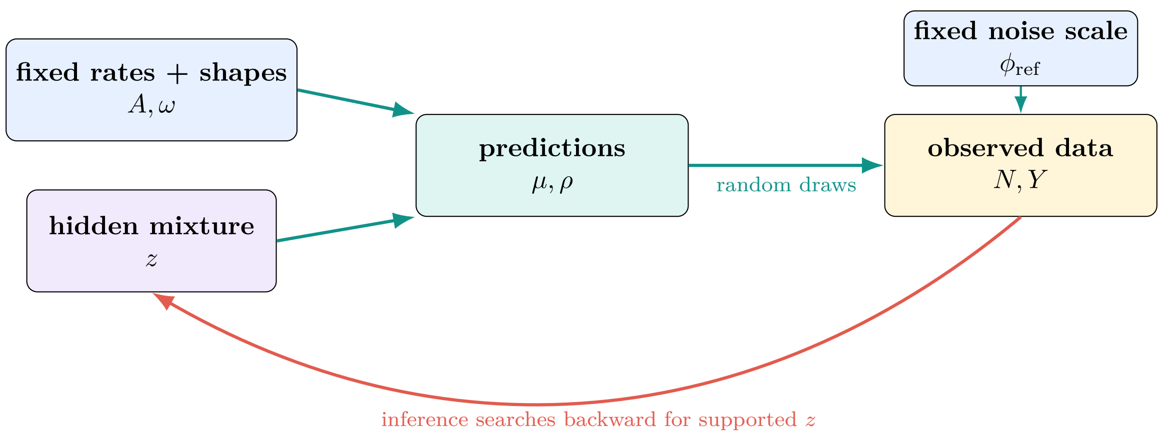


Figure 5: The model runs forward from a candidate mixture to possible data; MAP inference uses observed data to search backward.

What does "generative" mean?

A *generative model* is a mathematical recipe for how the model imagines that observable data could be produced. The forward, teal arrows in Figure 5 tell this story:

1. Start with one possible hidden mixture z.
2. Combine z with the fixed reference rates and shapes A, ω to calculate the predicted total μ and predicted bin fractions ρ.
3. Allow random experimental variation, controlled partly by ϕ_{ref}, to produce possible observed counts N, Y.

The boxes therefore have these jobs:

Box	Meaning
z	hidden effective cell-type amounts
A, ω	fixed reference information that turns a mixture into count and shape predictions
μ, ρ	predicted peak total and predicted short/mono/long fractions
ϕ_{ref}	fixed reference noise scale used in the negative-binomial draw
N, Y	total and length-bin counts actually observed

For the running example, one candidate mixture is

$$z_T = 1.5, \qquad z_B = 0.5.$$

Running this candidate *forward* through the model gives

$$\mu = 24, \qquad \rho = \left(\frac{10}{24}, \frac{8}{24}, \frac{6}{24}\right).$$

A possible observation is then

$$N = 24, \qquad Y = (10, 8, 6).$$

What does "inference searches backward" mean?

In a real experiment, the direction of the problem is reversed: we already know the observed N, Y, but we do not know the hidden z. ShapeMix therefore tries candidate values of z and asks:

> *If this were the mixture, how well would it explain the counts that we observed?*

For $N = 24$ and $Y = (10, 8, 6)$, three candidates with the same total effective amount predict different shapes:

T proportion	(z_T, z_B)	predicted short/mono/long	match
25%	$(0.5, 1.5)$	$(6, 8, 10)$	weak
50%	$(1.0, 1.0)$	$(8, 8, 8)$	intermediate
75%	$(1.5, 0.5)$	$(10, 8, 6)$	strongest

The software repeats the following search:

1. propose a possible z;
2. run that z forward to calculate μ, ρ;
3. score how well those predictions explain the observed N, Y;
4. include how plausible the candidate is under the Gamma prior; and
5. move toward a higher posterior score until it finds the MAP estimate.

Important distinction

The red arrow is not a claim that biology or time literally runs backward. It shows the direction of the computer's reasoning: from known observations toward an unknown explanation. A "supported z" is a candidate that explains the data well *and* is reasonably compatible with the prior. The MAP result is the highest-scoring point, not a confidence interval.

9.9 How one candidate z receives a likelihood score

For a proposed z, ShapeMix calculates the probability of the observed data in two parts:

$$\boxed{\underbrace{p(N, \boldsymbol{Y} \mid z)}_{\text{data likelihood}} = \underbrace{p(N \mid z)}_{\text{total-count score}} \times \underbrace{p(\boldsymbol{Y} \mid N, z)}_{\text{bin-split score}}.}$$

These are probability *masses*: they score the exact integer counts that were observed. The two factors answer different questions.

Part 1: score the observed total with a negative binomial

The candidate z determines e and the predicted mean μ. ShapeMix evaluates the negative-binomial probability at the observed total N:

$$p(N \mid z) = \text{NegBinomial}(N;\ \text{mean} = \mu,\ \text{inverse-dispersion} = \max(e, \epsilon)\phi_{\text{ref}}).$$

For the running candidate, $\mu = 24$, $e = 2$, and the tutorial uses $\phi_{\text{ref}} = 6$. The distribution is therefore centered at 24 with inverse-dispersion 12. Observing $N = 24$ receives a relatively high total-count score. A distant value such as $N = 60$ receives a much lower score, but not necessarily zero, because the model allows experimental noise.

Part 2: score the observed split with a multinomial

The same candidate predicts the fractions

$$\boldsymbol{\rho} = (\rho_{\text{short}}, \rho_{\text{mono}}, \rho_{\text{long}}).$$

Given that exactly N cuts were observed, the multinomial asks how plausible their allocation among the three bins was:

$$p(\boldsymbol{Y} \mid N, z) = \frac{N!}{Y_{\text{short}}!Y_{\text{mono}}!Y_{\text{long}}!} \prod_b \rho_b^{Y_b}.$$

For $N = 24$, $\boldsymbol{Y} = (10, 8, 6)$, and $\boldsymbol{\rho} = (10/24, 8/24, 6/24)$, this becomes

$$p((10, 8, 6) \mid N = 24, z) = \frac{24!}{10!\,8!\,6!} \left(\frac{10}{24}\right)^{10} \left(\frac{8}{24}\right)^{8} \left(\frac{6}{24}\right)^{6}.$$

The factorial factor counts all possible orders in which 10 short, 8 mono, and 6 long cuts could have appeared. This is like asking for the chance of seeing particular color totals after 24 random draws from three colored bins.

The candidate predicting $(10, 8, 6)$ receives a higher shape likelihood than candidates predicting $(8, 8, 8)$ or $(6, 8, 10)$. Exact equality is not a requirement: the multinomial assigns probabilities to many possible allocations.

Combine peaks, then add the prior

For many peaks, ShapeMix multiplies the two likelihood factors across peaks. In software it adds logarithms, which is numerically safer:

$$\log p(N, \boldsymbol{Y} \mid z) = \sum_p \left[\log p(N_p \mid z) + \log p(\boldsymbol{Y}_{p,\cdot} \mid N_p, z)\right].$$

The likelihood is the data-fit score. MAP then adds the Gamma-prior score:

$$\underbrace{\log p(N, \boldsymbol{Y} \mid z)}_{\text{fit to the observed data}} + \underbrace{\log p(z)}_{\text{prior plausibility}} = \text{posterior score, up to a constant.}$$

Important distinction

The likelihood $p(N, \boldsymbol{Y} \mid z)$ means "how probable were these data if this z were used?" It is not the probability that z is true. The posterior combines that likelihood with the prior. A candidate that exactly matches one peak is not automatically the MAP estimate: all peaks, count noise, and the Gamma prior contribute to the final score.

9.10 Count-only versus ShapeMix

Both arms use the same observed totals, reference signatures, prior, initialization, optimizer, seeds, and compute budget.

Term	Count-only	ShapeMix
negative-binomial total likelihood	yes	yes
conditional multinomial shape likelihood	no	yes
Gamma prior on z	yes	yes

The factorization in the preceding subsection does not double-count observations. It separates total-count information from allocation information.

Pause and check

If every cell type has exactly the same $\omega_{c,p,\cdot}$, can the shape term identify the mixture?
Answer: no. Then the predicted bin fractions do not change with z, so ShapeMix should agree with count-only within numerical tolerance.

10 Where the fixed reference quantities come from

You can understand and use the core model without memorizing this section's estimation formulas. Its purpose is to show why A, ω, and ϕ_{ref} are fixed before a held-out spot is analyzed.

10.1 The missing index: an individual reference cell

New symbol: i

Name: training-reference cell index.
Job: labels one individually measured, cell-type-labeled reference cell.
It is different from c: i identifies one cell; c identifies its type.

New symbol: $R_{i,p,b}$ and $T_{i,p}$

$R_{i,p,b}$ is the training-reference cut-site count for cell i, peak p, and bin b.
$T_{i,p} = \sum_b R_{i,p,b}$ is that reference cell's total at the peak.

If reference cell 17 is labeled T and $R_{17,42,\text{short}} = 3$, that cell contributed three short-parent cut sites to peak 42.

Let $\mathcal{I}_c$ denote the set of training-reference cells labeled as type c. Conceptually:

$$A_{c,p} \approx \text{average of } T_{i,p} \text{ over } i \in \mathcal{I}_c,$$
$$\omega_{c,p,b} \approx \frac{\text{bin-}b \text{ counts pooled over } i \in \mathcal{I}_c}{\text{all-bin counts pooled over } i \in \mathcal{I}_c}.$$

The actual estimates use frozen smoothing so sparse peaks never receive unstable zero probabilities. Exact formulas appear in Appendix A.

10.2 Why dispersion is cross-fitted

To estimate ϕ_{ref}, ShapeMix splits training-reference cells into two deterministic folds within each cell type. Each cell is compared with a mean learned from the opposite fold. This prevents a cell from helping construct the same mean used to measure its own deviation.

Main idea

Reference signatures, smoothing, peak selection, and dispersion use training reference cells only.
Held-out source cells and pseudo-spots cannot influence them.

11 What the software reports for each spot

The current MAP implementation emits:

Output	Contents	Interpretation
`abundance.csv`	$\widehat{z}_{s,c}$	effective, depth-scaled MAP abundances; not calibrated nuclei
`proportions.csv`	$\widehat{\pi}_{s,c}$	main composition estimate; each row sums to one
`diagnostics.json`	objectives, restarts, convergence and runtime information	fit diagnostics, not confidence intervals

For example:

Spot	T	B	Monocyte
spot 1	0.65	0.25	0.10
spot 2	0.15	0.70	0.15

These rows look like probability vectors mathematically, but scientifically they are estimated mixture fractions. A value of 0.65 does not mean "65% confidence." Spot-level credible intervals would require a new model version that approximates or samples the posterior rather than reporting only its maximum.

12 How the observed tensor is constructed

12.1 From a fragment row to cut-site counts

The primary input is a BGZF/tabix-indexed five-column 10x fragments file: chromosome, start, end, barcode, and `readSupport`.

1. Treat each row as one deduplicated fragment.
2. Ignore `readSupport` in the primary analysis so PCR support is not counted again.
3. Compute parent length $L = \texttt{end} - \texttt{start}$.
4. Give both endpoint cut sites the same parent-length bin.
5. Assign each endpoint independently to the unique selected, non-overlapping peak that contains it.
6. Store one sparse spot-by-peak matrix per length bin.
7. Require their elementwise sum to equal the ordinary peak-count matrix.

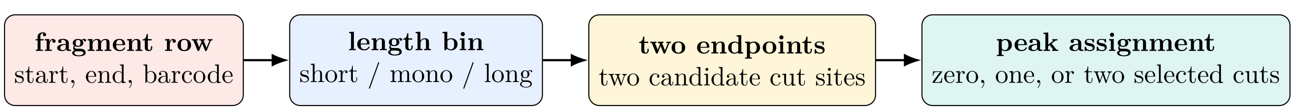


Figure 6: Each fragment is separated before its two endpoints are assigned; overlapping bars are never treated as one long fragment.

Version 1 selects 5,000 peaks by a deterministic accessibility-variance rank computed from reference cells only. Alternative peak counts and length cutoffs are versioned sensitivity analyses.

13 A fair experiment: isolate what shape adds

13.1 Leakage-free split

The primary PBMC source contains one donor. Each outer seed makes a deterministic, cell-type-stratified 70/30 split:

- the 70% training-reference pool selects peaks and estimates $A, \omega, \phi_{\text{ref}}$;
- the held-out 30% pool supplies cells for pseudo-spots; and
- inner mixture seeds are nested inside each outer split.

Both arms analyze the exact same pseudo-spots for every paired run. This design measures conditional resampling variability within one donor; it does not establish donor-level generalization.

13.2 Diagnostics and negative controls

After fitting, check:

- bin layers sum exactly to the ordinary peak totals;
- predicted totals μ reconstruct N reasonably;
- predicted bin fractions ρ reconstruct short/mono/long patterns;
- objectives, gradients, abundances, and proportions are finite;
- restarts converge to compatible solutions;
- homogenized ω removes ShapeMix's advantage; and
- permuted cell-type shape labels do not create a false improvement.

13.3 Primary comparison

The narrow claim is:

Main idea

Does the conditional parent-fragment length composition of the same observed cut sites improve cell-type proportion estimates beyond their peak totals?

Primary metrics compare $\widehat{\pi}$ with known pseudo-spot truth using RMSE and versioned Jensen–Shannon divergence. Runtime, memory, convergence, per-type errors, and rare-cell metrics are also reported even if shape does not help.

14 End-to-end worked example

We now repeat the entire running example without introducing any new symbols.

14.1 Step 1: fixed reference information

$$A_T = A_B = 12,$$
$$\boldsymbol{\omega}_T = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right), \qquad \boldsymbol{\omega}_B = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right).$$

14.2 Step 2: one candidate hidden mixture

$$z_T = 1.5, \qquad z_B = 0.5, \qquad e = 2.$$

Therefore:

$$\pi_T = 0.75, \qquad \pi_B = 0.25.$$

14.3 Step 3: predicted total

$$\mu = 1.5(12) + 0.5(12) = 24.$$

14.4 Step 4: predicted length split

$$\boldsymbol{v} = 1.5(12)\boldsymbol{\omega}_T + 0.5(12)\boldsymbol{\omega}_B = (10, 8, 6),$$
$$\boldsymbol{\rho} = \frac{(10,8,6)}{24} = \left(\frac{5}{12}, \frac{1}{3}, \frac{1}{4}\right).$$

14.5 Step 5: possible observation

$$N = 24, \qquad \boldsymbol{Y} = (10, 8, 6).$$

The observation happens to equal the prediction so the arithmetic is transparent; real observations fluctuate around their predictions.

14.6 Step 6: why shape identifies the recipe

Hold $e = 2$ fixed and compare three candidate T proportions:

π_T	(z_T, z_B)	predicted short/mono/long
25%	$(0.5, 1.5)$	$(6, 8, 10)$
50%	$(1.0, 1.0)$	$(8, 8, 8)$
75%	$(1.5, 0.5)$	$(10, 8, 6)$

All three candidates predict the same total, 24. The observed shape $(10, 8, 6)$ matches the 75% T candidate most closely.

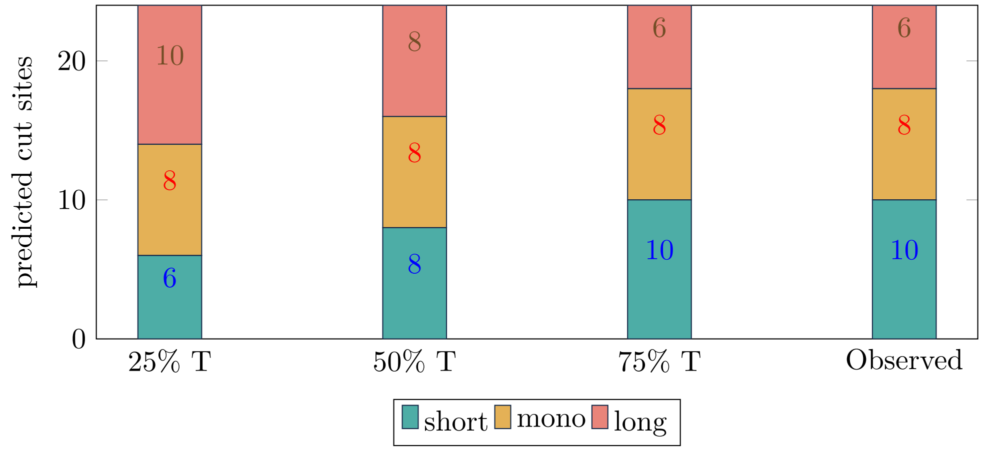


Figure 7: Peak totals are identical, but the length composition distinguishes the candidate recipes.

Important distinction

This exact match is a teaching calculation, not the complete MAP computation. The implemented result balances negative-binomial total fit, conditional multinomial shape fit, and the Gamma prior over continuous positive abundances.

15 Symbol glossary in the order learned

Table 1: Core symbols, ordered by first conceptual appearance.

Symbol	Meaning	Role
A	reference peak rate	fixed reference input
z	effective abundance	inferred by MAP
μ	predicted mean peak total	calculated
N	observed peak total	data
π	normalized cell-type proportion	reported output
$\boldsymbol{Y}$	observed vector of length-bin counts	data
$\boldsymbol{\omega}$	reference conditional length fingerprint	fixed reference input
$\boldsymbol{v}$	predicted length-bin counts	calculated
$\boldsymbol{\rho}$	predicted length-bin fractions	calculated
e	total effective abundance	calculated
ϕ_{ref}	reference inverse-dispersion	fixed reference input
p	selected-peak index	index
s	spatial-spot index	index
b	parent-length-bin index	index
c	cell-type index	index
ϵ	small positive numerical guard	fixed setting
i	individual training-reference cell index	index

Symbol	Meaning	Role
$R_{i,p,b}$	training-reference bin count	observed reference data
$T_{i,p}$	training-reference peak total	observed, calculated from R

Notation operators

Notation	Read it as
$\sum_c$	add over all cell types
$\sim$	is modeled as a random draw from
$\mid$	given
$\propto$	proportional to
$\widehat{z}$	estimated z
$\arg\max_z f(z)$	the value of z where f is highest
$\boldsymbol{Y}_{s,p,\cdot}$	the complete bin vector for one spot and peak

About primes on indices: a primed letter inside a sum is only a dummy counter. For example, $\sum_{b'}$ means "add over every bin," and $\sum_{c'}$ means "add over every cell type." It does not create a new kind of bin or cell type.

Pause and check

You understand the core ShapeMix model if you can explain these six lines in ordinary language:

$$v_{s,p,b} = \sum_c z_{s,c} A_{c,p} \omega_{c,p,b}, \quad \mu_{s,p} = \sum_b v_{s,p,b}, \quad \rho_{s,p,b} = \frac{v_{s,p,b}}{\mu_{s,p}}.$$

$$N_{s,p} \sim \text{NegBinomial}(\mu_{s,p}, \max(e_s, \epsilon)\phi_{\text{ref}}),$$

$$\boldsymbol{Y}_{s,p,\cdot} \mid N_{s,p} \sim \text{Multinomial}(N_{s,p}, \boldsymbol{\rho}_{s,p,\cdot}), \qquad \pi_{s,c} = \frac{z_{s,c}}{e_s}.$$

A Exact fixed-reference estimation

This appendix matches the executable specification. It is intentionally placed after the core model because its extra variables describe how fixed inputs are constructed, not what is inferred for a spot.

For cell type c, let $\mathcal{I}_c$ be its training-reference cells and let $h_i = 1$ be the same-protocol exposure. Define the pooled peak target:

$$g_p = \frac{\sum_i T_{i,p}}{\sum_i h_i}.$$

With $\alpha_A = 0.5$:

$$A_{c,p} = \frac{\sum_{i \in \mathcal{I}_c} T_{i,p} + \alpha_A g_p}{\sum_{i \in \mathcal{I}_c} h_i + \alpha_A}.$$

Define the dataset-global length distribution:

$$u_b^{\text{global}} = \frac{\sum_{i,p} R_{i,p,b}}{\sum_{i,p,b'} R_{i,p,b'}}.$$

With the frozen $\alpha_\omega = 1$, first smooth each pooled peak:

$$u_{p,b} = \frac{\sum_i R_{i,p,b} + \alpha_\omega u_b^{\text{global}}}{\sum_{i,b'} R_{i,p,b'} + \alpha_\omega},$$

then smooth each cell-type/peak fingerprint:

$$\omega_{c,p,b} = \frac{\sum_{i\in\mathcal{I}_c} R_{i,p,b} + \alpha_\omega u_{p,b}}{\sum_{i\in\mathcal{I}_c} T_{i,p} + \alpha_\omega}.$$

Say the equation in words

Sparse cell-type/peak observations borrow a small amount of information from the pooled peak pattern, which itself borrows from the dataset-global pattern. This keeps every conditional bin probability finite and positive.

A.1 Cross-fitted inverse-dispersion

For each reference cell i, estimate a held-out mean $m_{i,p}$ from the opposite within-type fold. Then:

$$\alpha_{\text{ref}}^{\text{raw}} = \frac{\sum_{i,p} \left[(T_{i,p} - m_{i,p})^2 - m_{i,p}\right]}{\sum_{i,p} m_{i,p}^2},$$

$$\alpha_{\text{ref}} = \max\left(\alpha_{\text{ref}}^{\text{raw}}, 10^{-8}\right), \qquad \phi_{\text{ref}} = \frac{1}{\alpha_{\text{ref}}}.$$

The fold seed, numerator, denominator, and result are recorded. Non-finite values or a zero denominator are errors.

B Optimization, implementation, and fixed contract

B.1 MAP fitting

Only z is inferred. The implementation uses

$$z = \text{softplus}(z^{\text{raw}}) + \epsilon$$

to keep abundances positive. It initializes from collapsed-count NNLS with a uniform fallback, uses deterministic restarts and Adam optimization, processes spots and peaks in chunks, and fails on non-finite losses, gradients, abundances, or proportions.

Advanced detail: objective comparison

$$\mathcal{L}_{\text{count}}(z) = \log p(N \mid z, A, \phi_{\text{ref}}) + \log p(z),$$
$$\mathcal{L}_{\text{shape}}(z) = \mathcal{L}_{\text{count}}(z) + \log p(Y \mid N, z, A, \omega).$$

Only the final conditional-shape term differs.

B.2 Sparse implementation invariants

- Reference and spatial objects have identical ordered selected peaks.
- The three ordered CSR layers are short, mono, and long.
- Every layer contains finite, nonnegative integers.
- The layer sum equals the ordinary peak-count matrix exactly.
- A zero-total spot/peak row contributes zero to the conditional term.
- Log probabilities use stable normalization; non-finite objectives or gradients are hard failures.

B.3 Fixed evaluation contract

Every prediction row must include the declared ordered cell-type universe, be finite and nonnegative, and sum to one within 10^{-6}. Missing declared types are penalized as zero; unknown extra types are errors. Invalid rows are never silently dropped or renormalized.

Primary metrics are:

- RMSE across all spot-by-type entries; and
- `jsd_v2`, the mean per-spot base-2 Jensen–Shannon divergence.

The one-donor repeated-split design quantifies conditional resampling variability, not donor-level or population-level uncertainty.

C Extensions that are not part of MVP version 1

The following may be valuable future versions, but none should be confused with the current output:

- posterior sampling or variational inference for approximate credible intervals;
- a Dirichlet–multinomial shape term for extra allocation variability;
- fixed or learned background signal;
- cross-protocol scaling;
- spatial smoothing;
- peak-specific dispersion;
- position, motif, or footprint shape axes; and
- learned rather than fixed reference signatures.

Each extension changes the model or scientific claim and requires a new version, negative controls, and an identifiability check.

D Summary in one page

Main idea

ShapeMix asks whether the parent-fragment length composition of ATAC cut sites improves deconvolution beyond the same cut sites collapsed to peak totals.

1. A spot has hidden positive effective cell-type amounts z.
2. Fixed reference rates A predict how many cut sites each type contributes.
3. Fixed reference shapes ω predict how those cuts divide among short, mono, and long parent-fragment bins.
4. A negative binomial models each noisy peak total N.
5. A conditional multinomial models how the same total divides into Y.
6. A fixed $\mathrm{Gamma}(2, 1)$ prior regularizes every z.
7. MAP reports one best abundance vector, normalized to proportions π.
8. Version 1 reports no spot-level credible or confidence intervals.

$$\boxed{v_{s,p,b} = \sum_c z_{s,c} A_{c,p} \omega_{c,p,b}, \qquad \rho_{s,p,b} = \frac{v_{s,p,b}}{\sum_{b'} v_{s,p,b'}}}$$

$$\boxed{N_{s,p} \sim \mathrm{NegBinomial}\left(\sum_b v_{s,p,b}, \max(e_s, \epsilon)\phi_{\mathrm{ref}}\right), \quad \boldsymbol{Y}_{s,p,\cdot} \mid N_{s,p} \sim \mathrm{Multinomial}(N_{s,p}, \boldsymbol{\rho}_{s,p,\cdot})}$$

$$\boxed{z_{s,c} \sim \mathrm{Gamma}(2, 1), \qquad \pi_{s,c} = \frac{z_{s,c}}{\sum_{c'} z_{s,c'}}.}$$

References and positioning notes

References

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